

Chapter 17

Production of Magnesium and Magnesium Alloy Powders

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INTRODUCTION

Powders and granules of magnesium and their alloys find various applications in modern technology, such as modifying metallurgical properties of metals; manufacturing Grignard reagents [1] that are used in organic synthesis to produce pharmaceuticals, perfumes, and other fine chemicals; chemical reduction; additives in electric welding electrode flux; and manufacture of corrosion-resistant constructional purpose details.

Magnesium-based alloys are used in various applications in the aerospace, automotive, and 3C (computer, communication, and consumer electronics products) industries. These applications are driven largely because of a combination of low density, low melting point, good castability, and high specific stiffness of magnesium alloys. Further, high-strength, magnesium-based alloys also have the potential to be used as lightweight armor, which can provide superior protection against modern threats [2]. These applications demand high strength and high ductility that cannot be achieved by the traditional casting process.

Powder metallurgy offers a unique route to targeting novel compositions and unique microstructures leading to high-performance lightweight magnesium applications. Rapidly solidified powder and the use of nanostructured powder have been studied and reported in the literature with promising results [3–7].

Magnesium is a reactive material like aluminum and titanium (Table 17.1). The heat of formation, ΔH_0 , of magnesium oxide equals of 603.2 kJ/mol. Magnesium is potentially explosive in the powder form. The lower explosive limit of the powder with particles smaller than 74 μm amounts to 10 g/m³ and the rate of pressure rise equals 6.3 MPa/s (see Chapter 27). The development of the production methods should take into account these features of magnesium.

PRODUCTION METHODS

The basic methods of magnesium powder production are mechanical crushing (scratching of an ingot from magnesium card tape fixed on a rotating drum, milling of an ingot), atomization of molten metal, evaporation-condensation, and electrolysis.

Mechanical Crushing

The first industrial production of magnesium powder originated for pyrotechnic purposes in the early 20th century in Germany. The method consists of scratching a magnesium ingot by a card tape fastened on a rotating drum. The machine tool intended for this process, the “kratz machine,” has been improved for a long time. Powder dispersivity is adjusted by the speed of magnesium plate feeding, the speed of drum rotation, and the diameter of the card tape needles. The drum

TABLE 17.1 Properties of Reactivity of Mg in Comparison With Al and Ti

Material	Density (g/cm³)	Melting Point (Approximate) (K)	Boiling Point (K)	Heat of Fusion		Specific Heat Capacity		Combustion ^a	Heats of Formation Oxides			
				kJ/ mol	cal/ mol	J/ g·K	cal/ g·K		K	Oxide	Heat of Formation (ΔH ₀) ^b	
											kJ/mol	cal/mol
Magnesium	1.74	922	1363	8.954	2.139	0.90	0.215	3373	MgO	−603.2	−144.090	
Aluminum	2.70	933	2740	10.790	2.577	1.02	0.244	3773	Al ₂ O ₃	−1691.5	−404.080	
Titanium	4.51	1993	3560	15.450	3.690	0.52	0.124	3273	TiO ₂	−955.9	−228.360	

^aSource: Ref. [7].
^bThe adiabatic combustion temperature in oxygen at 100kPa pressure.

moved along its axis to enable the entire plane of the plate to be processed uniformly. Abrasion uniformity is provided by a continuous back and forth motion of the plate, which entered into the case of the machine tool through a special aperture. The card tape is reeled on a steel drum. The diameter of the drum is commonly 200 mm; the needle speed on the surface is about 1000 m/min. The thickness of the magnesium plate is usually 40 mm by 350 mm width. The minimal diameter of the card tape needles was 0.64 mm. To prevent sparking while the needles are falling out from the tape, the case of the kratz machine is manufactured of aluminum-magnesium alloy. The received powder gets into a box, which is installed in the casing of the kratz-machine. The finest magnesium powders – intangible by touch – were produced by additional crushing of the powder manufactured in the kratz machine in ball mills in the atmosphere of carbon dioxide.

Magnesium chips and grit are also produced by cutting cylindrical blanks 200–300 mm in diameter in a lathe using a special sawtooth cutter. The average particle size is 1 mm; the content of particles smaller than 0.75 mm in size is not higher than 35 wt%. The capacity of the machine tool is 8.8–15 kg/h.

Along with other methods, a method to produce magnesium powders by milling an ingot with subsequent crushing of the chips in a grinding machine and a hammer mill is applied in the United States. This method is more efficient than the above method of producing powders with the help of the card tapes. Special machine tools equipped with vertical and horizontal cutters are used. The produced chip-shaped powder is sucked into the collecting system, precipitates in a cyclone, and is divided by a screening machine to commodity fractions by particle size. The smallest fractions are caught by an oil filter placed in front of a ventilator. When necessary, powders are exposed to rolling in a ball mill for obtaining particles with a spherical shape (Fig. 17.1).

Characteristics of the powders produced by the method of mechanical grinding are given in Table 17.2. These powders commonly contain not <99.5% Mg and not >0.05% Fe, 0.005% Cl, and 0.1% of moisture (in weight).

Melt Atomization

Magnesium powders produced by inert gas atomization are polydisperse and have a particle size from several microns up to 0.5–1 mm. The hardware for the atomization is similar to that used for producing aluminum powders (see Chapter 15). Powders are classified into required fraction particle sizes.

The technology developed in VAMI (the Russian National Aluminum-Magnesium Institute) consists of atomization of the melt by jets of nitrogen with an oxygen additive of 3 ± 0.1 wt% while the melt temperature was 50–70°C higher than the melting point of magnesium or its alloy [8]. An optimum relation between the temperature of the melt and the atomizing gas, the pressure of the gas, the ratio of gas to metal mass flow, and other parameters as well as the optimum hardware-technological circuit of production provide fire and explosion safety for manufacture. Magnesium powders with the particle size smaller than 200 µm ranging from spherical (particles smaller than 50–100 µm) to oval (larger particles) are produced by this technology.

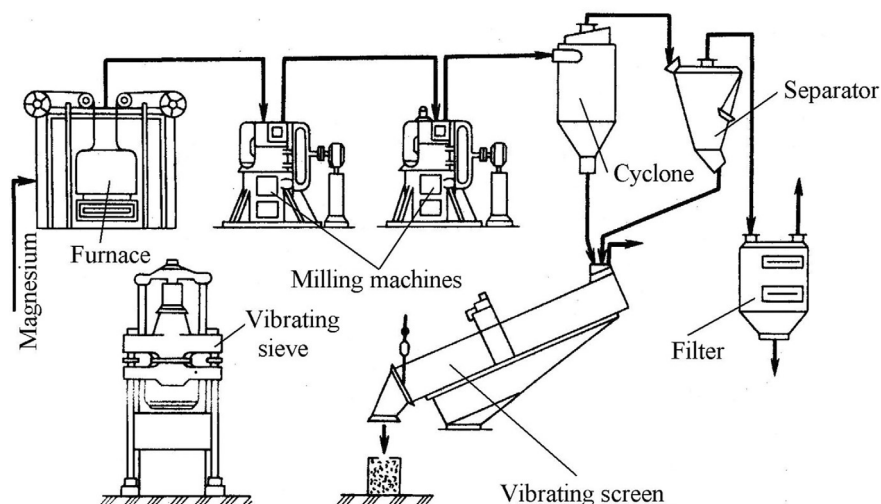


FIG. 17.1 The hardware-technological circuit of magnesium powder production.

TABLE 17.2 Properties of Typical Commercial Grades of Magnesium Powders

	Chemical Requirements (wt%) Max			Apparent Density (g/cm ³)	Particle Size Distribution (μm)	Source
Type	Mg	Alloying Elements	Admixture			
Turnings and chips						
LNR 11	99.5 min	...	...	0.4	1000–3000	ECKA
LNR 15	99.5 min	...	...	0.5	300–600	ECKA
LNR 16	99.5 min	...	...	0.6	3000–4000	ECKA
AZ91D	90	9Al-1.0Zn	...	0.6	1000–3000	ESKA
MG Turnings/chips	...	...	...	...	(3–6 mm) × (<1 mm) × (5–15 mm) (<2 mm) × (1 mm) × (<8 mm)	JLM
MG Alloy scrap	...	...	...		+2.8 mm <2% –0.85 mm <5%	JLM
Powders						
PF 01/91	99 min	...	...	0.9	315–630	ECKA
PF 91/51 Type III	99.5 min	...	...	0.7	100–315	ECKA
PK 31 Type IV	97 min	...	...	0.5	<71	ECKA
Pure magnesium	99.8		Cu (0.02% max) Pb (0.01% max) Mn (0.1% max) Ni (0.001% max) Na (0.006% max) Sn (0.01% max)		A: +200 mesh: 0.5% max –325 mesh: 60% max B: +200 mesh: 0% max –325 mesh: 95% min	MEP
Electron MAP+21	Bal.	2.7Nd, 1.3Gd, 0.3Zn, 0.6Zr/wt%	...	...	A: <150 μm; d ₅₀ =90 μm B: <25 μm	MEP
Electron MAP+43	Bal.	4Y, 3Nd, 0.5Zr/wt%	...			ME
Electron MAP+91	Bal.	(8.5–9.5)Al, (0.45–0.9)Zn, (0.17–0.3)Mn/wt%				MEP
Electron MAP 21	Bal.	2.7Nd, 1.3Gd, 0.3Zn, 0.6Zr/wt%	...	...	(A) 80 × 325 size: +80 mesh: 5% max –325 mesh: 20% max	MEP
Electron MAP 43	Bal.	2.7Nd, 1.3Gd, 0.3Zn, 0.6Zr/wt%	...	...	(D) 325 × D size: +325 mesh: 5% max –325 mesh: balance	MEP
Electron MAP 91	Bal.	(8.5–9.5)Al, (0.45–0.9)Zn, (0.17–0.3)Mn/wt%	...	...		MEP

Electron 21	Bal.	2.7Nd, 1.3Gd, 0.3Zn, 0.6Zr/wt%	...	...	–80+325 mesh size: +325 mesh: 5% max	MEP
AZ31 B	Bal.	3Al, 1Zn, 0.3Mn	...	...		MEP
AZ91 E	Bal.	(8.5–9.5)Al, (0.45–0.9) Zn, (0.17–0.3)Mn/wt%	...	...		MEP
WE43B	Bal.	4Y, 3Nd, 0.5Zr/wt%	...	...		
MG Powder	...	...	...	...	18–80 mesh (1.0–0.6 mm); 40–90 mesh (0.425–0.17 mm)	JLM
Passivation MG powder	...	...	...	...		JLM

Companies producing the commercial powders: ECKA, Eckart-Werke (trading companies: Eckart Austria and Eckart Switzerland); MEP, Hard Metals Inc. (Magnesium Electron Powders, PA, United States); Niagara Metallurgical Products Ltd. (Canada); JLM-Hebi City, Jiang Lang Metal Co. Ltd. (China).

Production of Magnesium Granules

Granulated magnesium is produced by atomization of the melt with the use of a salt flux for the protection of the metal against oxidation [8]. A flowchart of the spinning cup granulation process is represented in Fig. 17.2. The atomizing chamber, in which the magnesium melt atomization, cooling, and collection of the formed granules are carried out, has an internal diameter of 3.2 m. In the center of the chamber, a spinning cup with punched walls is placed; its rotation speed is in the range from 800 to 1600 rpm. The melted magnesium is introduced from the furnace into the spinning cup by a siphon. Simultaneously, the salt flux protecting magnesium from oxidation is brought into the spinning cup.

The flying out melt droplets through punched spinning cup walls under centrifugal force into the chamber are solidified in flight before colliding with chamber walls. With the help of a screw and pneumatic transport, the produced granules are removed from the chamber and sieved into size fractions. The fraction of 1.6–0.5 mm is a commercially useful fraction, the yield of which is higher than 90%–95%. The apparent density of granules is 0.85–0.95 g/cm³. The granules consist of 90%–95% spheroidal particles while the content of the needle-shaped ones must not exceed 10% because, by their greater content, the flow rate deteriorates. A micrograph of spheroidal and needle-shaped magnesium granule particles is represented in Fig. 17.3. The capacity of the device is of 8000–10,000 tonnes/year.

Granules contain from 80 to 96 wt% of active magnesium while the rest is an oxide-salt film that can be separated from metal particles by abrasion, screening, or dissolution of the crushed salt phase. Afterward, the content of active magnesium in granules achieves 98.5–99.5 wt%. Application of such granulated magnesium is effective for desulfurization and modifying of pig iron, thermochemical and thermoacid processing of oil wells in the oil-extracting industry, and magnesium-organic synthesis of polyorganosiloxanes.

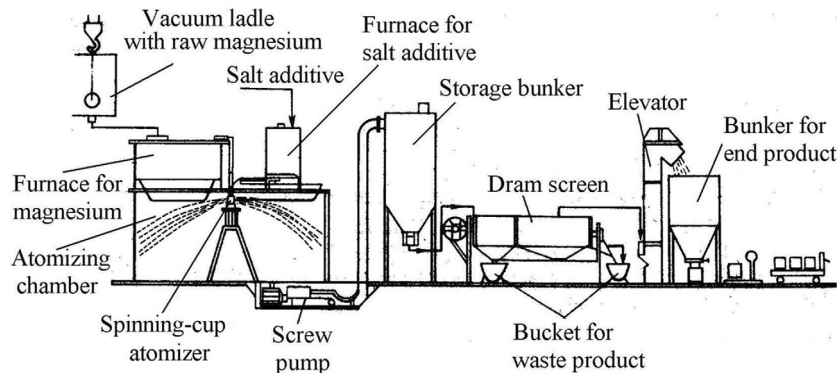


FIG. 17.2 Flowchart of the spinning cup magnesium granulation process.

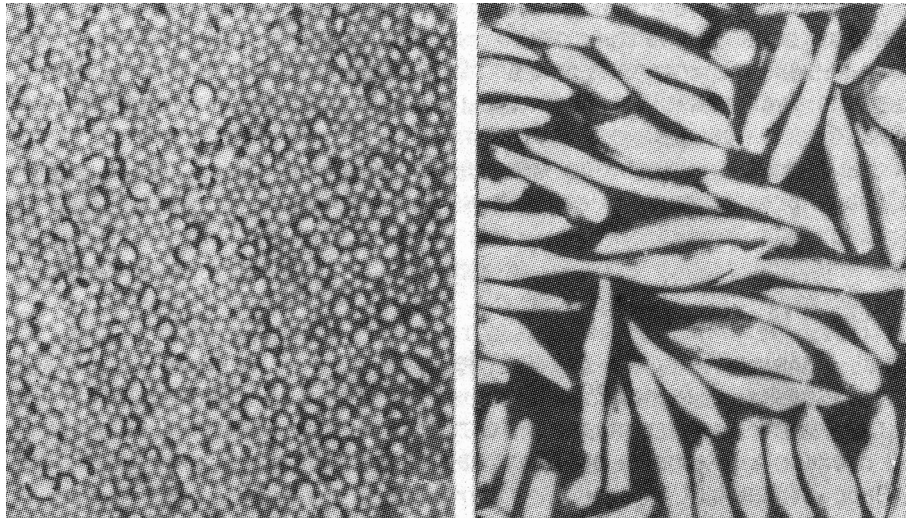


FIG. 17.3 Micrograph of spheroidal and needle-shaped magnesium granule particles with various magnification.

In Russia, devices for producing large granules (5–20 mm in diameter) by the atomization of molten magnesium in a fusible mix of alkaline chlorides and alkaline earth metals are used. The molten magnesium is atomized on the surface of a fusible salt melt at a temperature of 723–773 K. The metal drops solidify into granules from 3 to 25 mm in size and emerge on the melt surface, where, with the help of a scoop with a perforated bottom, they are separated from the salt melt and thrown into a rotating cooler drum. Granules at a temperature of 473–523 K are fed into a drum-type screening device for the separation of the commodity fraction (5–20 mm in diameter). The commodity fraction contains 1.5%–2.0% of salts.

The productivity of the device as a saleable fraction is from 1200 to 1500 kg/h with a yield of 90%–92%. It was found that the temperature of the molten salt (the temperature of the melted magnesium was maintained automatically in the interval 980 ± 10 K) exerts a principal influence on the saleable fraction yield. Its maximum is observed at temperatures of 773–823 K; when the salt melt temperature is raised to 873 K, the fraction of >15 mm consists almost completely of nonspherical particles. For manufacturing granules smaller in size, the installation with the spinning cup running at rotational speeds from 3000 to 10,000 rpm is used [9].

In the VSMPO-AVISMA Corporation (or metal-producing company of Verkhnyaya Salda, Russia), a granulated magnesium technique was developed consisting of magnesium melt or its alloys together with the salt mix and insertion into the melt of a dispersing agent (silicon oxide) with active stirring of the melt for metal dispersion. Then the melt is cooled to a temperature below the melting point of the metal, and granules are taken from the melt. The extracted granules are separated from solid chloride salts by screening. The granulated magnesium particles are roundish. The material is not caked and possesses good flowability that provides good pneumatic transport in the technological process. Granules are also produced from alloys of magnesium alloyed by aluminum, zinc, titanium, and other elements.

Coarse spherical granules of magnesium are produced by the technology based on the disintegration of liquid-metal magnesium jets in a molten salt. Molten metal in the form of the jets of the required diameter moving with a set speed is inserted into the salt melt, the temperature of which is lower than the temperature of magnesium crystallization. Fluctuations of internal pressure are created in metal jets with the help of the magneto-hydrodynamic method (MGD), and they are transformed into fluctuations of the jet surface, resulting in their disintegration into drops of equal size. Drops are cooled to ambient temperature and separated from molten salt as granules.

The technology allows the production of granules of magnesium and magnesium alloys of predetermined size in the range from 3 to 30 mm in diameter. The obtained granules are characterized by a dense structure; they contain up to 98% metallic magnesium and not $>2\%$ chloride. Apparent density of the product is 1.1 g/m^3 .

Coarse spherical granules of magnesium of 5–15 mm in diameter are used for the production of organosilicon compounds by the method of continuous magnesium-organic synthesis. It is necessary to use large and heavy granules because granules of smaller sizes and thus a smaller weight are carried away with the products of synthesis. This increases the explosion hazard of the process due to secondary reactions with the formation of hydrogen. Such granules have a dense structure and a high content of the basic component – magnesium.

A method of producing fine spherical granules of magnesium was developed in the VSMPO-AVISMA Corporation. This method is based on capillary disintegration of magnesium jets under the action of regular fluctuations of the pressure created in the metal by the MGD method. In this technique, the inert atmosphere is used only during jet disintegration and drop spheroidization. Further, the drops get into the air atmosphere where they are finally crystallized and cooled down to the required temperature. The method is economical and does not demand a hermetic chamber. The method produces granules of the narrow size between 0.3 and 30 mm in diameter. Granules have a shining metal surface and a spherical shape. The apparent density of the granules is $1.0\text{--}1.1 \text{ g/cm}^3$. The content of metallic magnesium is 99.7% min.

A magnetic field facilitates a refining of a granule structure and hence expands the area of their usage not only as a chemical reagent (ferrous and nonferrous metallurgy, chemical industry, etc.), but also in powder metallurgy for producing high-strength materials by virtue of their low density. Such products are potentially suitable for mechanical engineering applications, including aviation and space technology.

The VSMPO-AVISMA Corporation developed a method of atomizing the molten magnesium by a magnetodynamic method in air. It is based on the discovery that magnesium droplets do not ignite in a strong magnetic field. The technology allows the production of monodisperse granules of equal size, shape, and weight with average characteristics varying within the ranges: weight of one granule, from 0.1 to 20 mg; size of granules (diameter of the head part), 0.3–3.0 mm; length of the granule, 1.5–15 mm; apparent density, $0.3\text{--}0.9 \text{ g/cm}^3$; content of metallic magnesium, $>99.5\%$ or $>99.8\%$. The technology is used at the Solikamsk magnesium factory.

Gas-Phase Method

A high steam tension of magnesium vapor allows the use of the method of magnesium evaporation and vapor condensation for the production of fine magnesium powders. Two known types of gas-phase precipitation, physical vapor deposition

(PVD) and chemical vapor deposition (CVD), are described in Chapter 5. The PVD process includes melting the metal, vaporization of the melt, feeding the vapor into the zone of condensation, condensing the vapor into powder either in the free space or on a cooled surface (in a static or moving gas medium), and accumulating the powder in the cold part of the volume. The process is carried out in the presence of a neutral gas, sometimes with the addition of an oxidant or, on the contrary, of a passivating agent. The evaporation of the material is performed by an electron beam or a magnetron sputtering laser, either in a resistance or induction furnace.

Frishberg et al. reported about magnesium powder production by the gas-phase method on installation utilizing an induction furnace for the metal evaporation [10]. The vapor condensation in this installation is carried out in vacuum or in a rarefied atmosphere of inert gases. Vapor is condensed on a cooled brass drum 220 mm in diameter with its rotation speed being controlled; a knife cuts off the condensate. A graphite crucible containing the melt is closed by a cover with an adjustable tapping. Mixing vapor with the inert gas takes place in a special branch pipe. An end face of a branch pipe closely approaches the rotating surface of a crystallizer. Heating of the crucible is inductive. For producing magnesium powder with a particle size smaller than 1 μm , the temperature of the melt was 993 K, the gas pressure was 13.3 Pa, and the rotation frequency of the condenser was 0.75 min^{-1} . Key process parameters consist of the temperature of metal evaporation and condensation, the nature and pressure of the inert gas, and the method and rate of condensate removal from the condensation zone.

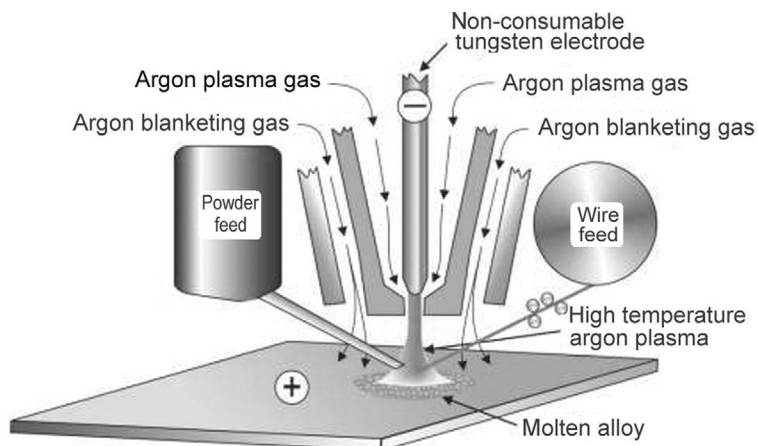
Electrolysis

Crystalline magnesium in powder condition is precipitated on a cathode by electrolysis of the fusible melts down the magnesium melting point [8]. The melts contain magnesium chloride such as KCl-MgCl₂, KCl-NaCl-MgCl₂, KCl-Li; KCl-NaCl-BaCl₂, and MgCl₂ are usually used in the capacity of electrolytes. Soluble (magnesium, magnesium-based alloys) and insoluble (graphite) anodes are applied. The melt temperature is in the 673–793 K range, the current density on the anode is in the 0.1–0.5 A/cm² range, and on the cathode is in the 0.4–2.0 A/cm² range. The current yield reached 90%. The electrolytic magnesium powders are characterized by dendrite and acicular shapes and form the aggregates of 100–300 μm size while the sizes of separate crystals are from 5 to 60 μm . The dispersivity of powder particles is reduced with increasing temperature and duration of electrolysis and rises with current density increasing. The powders differ by high chemical purity.

Plasma-Transferred Arc Melting Process

For the purpose of improvement in alloying magnesium base alloys, Withers et al. applied a melting process with utilization of a plasma transferred arc (PTA), producing a mini melt pool [11]. The authors reported that mini pools of a 10–50 g size are small enough with rapid cooling to prevent density segregation and loss by volatilization. A PTA arc generates considerable stirring and oscillation in the mini pool that provides immediate solution of the additive alloying element. An illustration of a PTA torch that generates mini pools in the general size range of 10–20 mm diameter and 5–15 mm deep is shown in Fig. 17.4. An operating PTA system includes powder and wire feeders to add alloying elements to the basic mini melt pool of Mg, Al, or Ti.

FIG. 17.4 Schematic view of a plasma transferred arc (PTA) torch that produces a mini melt pool.



The alloy of nonequilibrium and immiscible constituents such as magnesium and titanium was synthesized as 50Mg-50Ti by PTA mini pool processing. This alloy showed a flexure strength of approximately 500MPa.

APPLICATIONS

Powders and granules of magnesium and their alloys find various applications in modern technology, including:

- Modifying metallurgical properties of metals, such as in the production of wrought steel and the removal of sulfur from hot iron products of blast furnaces.
- Manufacture of Grignard reagents, which are organometallic halides such as ethyl magnesium chloride (C_2H_5MgCl) that are used in organic synthesis to produce pharmaceuticals, perfumes, and other fine chemicals.
- Chemical reduction, as in the manufacture of beryllium and uranium.
- Additives in an electric welding electrode flux.
- Action as a light source in flares and photoflash bombs.
- Production of constructional purpose details in cases where only magnesium alloy powders and granules are applied.
- As an initial material for producing constructional products by the technique of the so-called thixotropic shaping, a new version of casting under pressure.

Nowadays, one of the basic consumers of granulated magnesium is ferrous metallurgy, where granulated magnesium in the pure state or as a mixture with lime and calcium carbide is applied for out-of-furnace pig-iron desulfurization. Granulated magnesium with a salt coating is applied in the pure state due to fire and explosion safety that provides the maximum effect.

Granulated magnesium is also applied in the structure of the so-called “powder” wire used for desulfurization of pig-iron by insertion of a wire into a ladle of melted pig-iron. Granulated magnesium with a salt coating is used in the production of high-strength spheroidal graphite cast iron.

Granulated magnesium with salt additives can be effectively used as a source of hydrogen production.

Granulated magnesium of various coarseness (from 1 to 20mm) is used in the development of effective technologies for the synthesis of silicon-organic monomers.

There is expected to be a consumption growth for fine granules of magnesium alloys in the production of computers, mobile phones, and in the motor industry, where manufacture of thin-walled details by the method of thixomoulding is required. The efficiency of this method significantly depends on the method of granule production as well as their structure and size.

Spheroidal magnesium alloy powders of the three compositions (Table 17.2) for additive manufacturing technologies (trade name: Electron MAP+) are manufactured by HARD METALS Inc. (Magnesium Electron Powders). These powders are inert gas atomized and typically available in particle sizes as: (A) $<150\mu m$ ($d_{50}=90\mu m$) and (B) $<50\mu m$ ($d_{50}=25\mu m$). A micrograph of characteristic Electron MAP+ powder particles is represented in Fig. 17.5.

These powders are intended for use by selective laser sintering, direct laser sintering, electron beam melting, direct metal deposition, laser engineering net shapes, cold spray deposition, and other PM technologies.

The gas atomized powders (trade name: Electron MAP) with compositions such as Electron MAP+ powders are available in particle size as: (A) 80×325 size: +80 mesh ($177\mu m$) of 5% max, -325 mesh ($44\mu m$) of 20% max, and

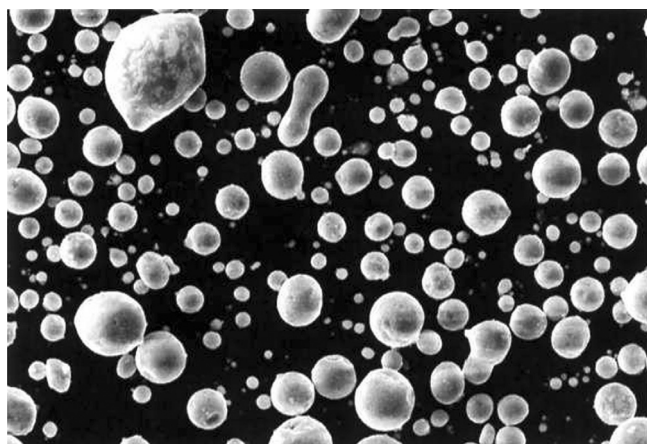
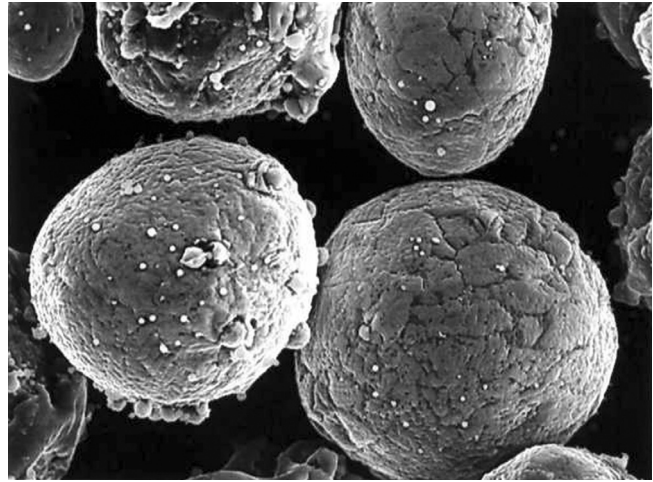


FIG. 17.5 Micrograph of characteristic Electron MAP+ magnesium alloy powder particles. (Courtesy of Magnesium Electron Comp.)

FIG. 17.6 Micrograph of Electron MAP magnesium alloy powder particles. (Courtesy of Magnesium Electron Comp.)

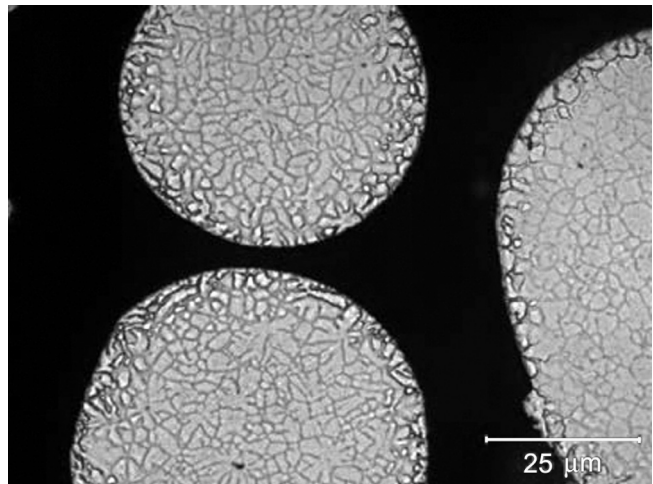


(B) 325 × D size: +325 mesh of 5% max, −325 mesh—balance. Fig. 17.6 illustrates the morphology of Electron MAP powder particles. The Electron MAP powders are assigned for use by powder forging, powder extrusion, hot and cold isostatic pressing, and equal channel angular extrusion.

Magnesium alloy powders of Elektron MAP43 (4 wt% Y, 3 wt% Nd, 0.5 wt% Zr, balance – Mg) and Elektron MAP 21 (2.7 wt% Nd, 1.3 wt% Gd, 0.3 wt% Zn, 0.6 wt% Zr, balance – Mg) were produced by Tandon and Madan [12] using the inert gas atomization process. The powders showed a fine dendritic microstructure (Fig. 17.7) resulting from rapid solidification due to the atomization process by a scalable gas atomizer [13]. The interdendritic spacing was between 1 and 3 μm for the −100/+325 mesh (150/44 μm) size. Apparent density, flow rate, and median diameter are for Electron MAP 43: 0.99 g/cm³, 65 s/50 g, and 83 μm and for Electron MAP, 21: 0.99 g/cm³, 65 s/50 g, and 83 μm, respectively. As the authors reported during the forging process, the powder was cold isostatically pressed into a preform and then subsequently heated and placed in a granular pressure transmitting media. A rapid application of large uniaxial pressure along with a smaller radial pressure resulted in rapid densification and full densification. The yield strength of the PM extruded Elektron MAP 43 alloy was 260 MPa with a UTS of 290 MPa, and an elongation of approximately 20%, which compared favorably against commercially extruded cast Elektron 43 because the latter is characterized by yield strength of 170 MPa with a UTS of 220 MPa, and an elongation of approximately 2% [14].

The PM consolidated atomized Elektron MAP 21 showed higher strength as the yield strength of 350 MPa with a UTS of 360 MPa and an elongation of 10% while the commercial cast extruded Elektron 21 is characterized by a yield strength of 225 MPa with a UTS of 260 MPa, and an elongation of approximately 17% [15].

FIG. 17.7 Optical micrograph of Elektron MAP 43 size fraction −100/+325 mesh (150/44 μm). (From Tandon R, et al. *Influence of powder characteristics on the consolidation behavior of magnesium alloys for structural and energetic applications*, In: *Proceedings of the 2012 international conference on powder metallurgy and particulate materials*. Basel (Switzerland): Metal Powder Industries Federation; 2012. p. 07-157–07-163.)



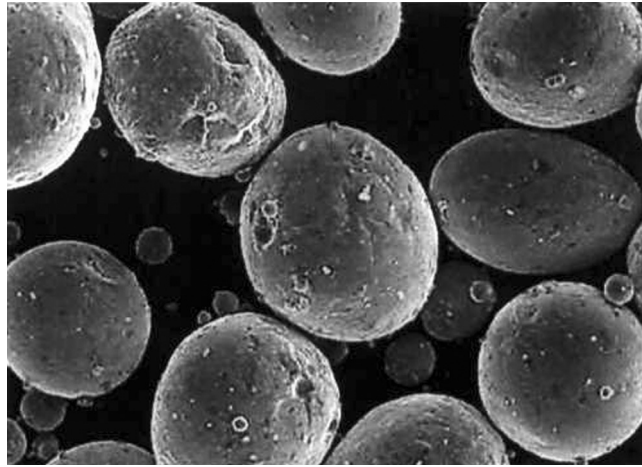


FIG. 17.8 Micrograph of Magnesium Electron pure magnesium powder. (Courtesy of Magnesium Electron Comp.)

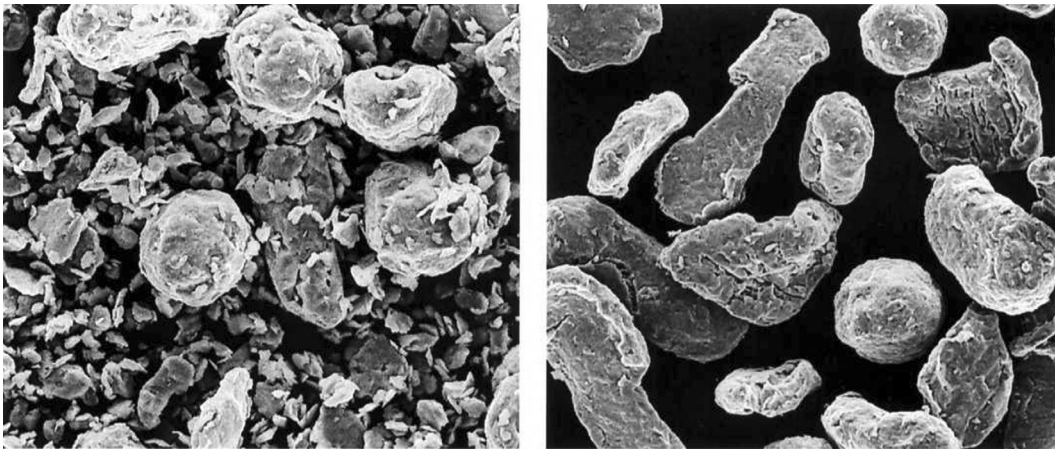


FIG. 17.9 Coarse and fine ground magnesium alloy powders of Magnesium Electron Comp. (Courtesy of Magnesium Electron Comp.)

The pure magnesium powders are made by the Magnesium Electron Company as blending additives for powder metallurgy parts and are available in particle size as: (A): +200 mesh (74 μm) of 0.5% max, –325 mesh (44 μm) of 60% min and (B): +200 mesh of 5% max, –325 mesh of 95% min). The powders are spheroidal without satellites (Fig. 17.8).

Coarse and fine ground magnesium alloy powders of the three compositions (Table 17.2) are offered by the Magnesium Electron Company for PM applications. The particles of these powders have an irregular shape and uneven surface (Fig. 17.9).

Nanocrystalline $\text{Mg}_{97}\text{Zn}_1\text{Y}_2$ (at%) alloys having high specific tensile strength, high elevated temperature tensile strength, high plasticity, and using a rapidly solidified (RS) powder metallurgy process were developed by Kavamura and Inoue [3]. The RS raw material was prepared in ribbon form by a single roller melt spinning method at a circumferential velocity of 42 m/s. The RS ribbons were then annealed in a vacuum at 573 and 673 K for 1.2 ks. The RS powder metallurgy process was realized by a closed processing system in an argon gas (O_2 , H_2O < 0.5 ppm) atmosphere on the laboratory installation. The details of the processing system are described in the article [16]. The RS $\text{Mg}_{97}\text{Zn}_1\text{Y}_2$ powder was produced by He-gas atomization at a gas-pressure of 9.8 MPa, and at a molten-alloy temperature of 1048 K. The powder, which was sieved to <32 μm , was used for the subsequent consolidation. The cooling rate of the sieved powder was estimated to be $1 \times 10^5 \text{ K}\cdot\text{s}^{-1}$ or more. The sieved powder was first cold pressed in a copper can with an inner diameter of 20 mm and an outer diameter of 23 mm, and then degassed for 900 s at room temperature. To consolidate the RS powder, extrusion of the powder was performed at an extrusion ratio of 10, at temperatures ranging from 573 to 723 K, and at a ram speed of 2.5 mm/s.

The authors [3] reported that the RS $\text{Mg}_{97}\text{Zn}_1\text{Y}_2$ powder and its PM alloy consolidated at 573 K predominantly consist of a hexagonal closed packed (hcp) Mg phase. The alloy mainly consists of hcp-Mg grains with and without a high density

of plane faults. The grain size was estimated to be about 180 nm. A small amount of fine rectangular Mg_{24}Y_5 precipitates with a grain size of about 7 nm was observed in the matrix grains.

The dimensions of the tensile specimens were a diameter of 2.5 mm and a gauge length of 16 mm for ambient temperature tests, and a diameter of 1.5 mm and a gauge length of 10 mm for elevated temperature tests [3]. As reported, the yield strength ($\sigma_{0.2}$) decreased from 610 to 430 MPa and the elongation increased from 5% to 16% with an increase in the extrusion temperature from 573 to 723 K. The Young's modulus was 45 GPa. The density of the alloy was 1.84 g/cm³. The RS PM alloy extruded at 573 K exhibited higher yield strength (YS) of 610 MPa and an elongation of 5%. The specific tensile strength ($\sigma_{0.2}/\rho_m$, where ρ_m is the density of the alloy) was about four times as high as that of the commercial AZ91-T6 I/M alloys (83 MPa/ ρ_m).

The study of the testing temperature dependence of the mechanical properties of the RS PM $\text{Mg}_{97}\text{Zn}_1\text{Y}_2$ alloys consolidated at 573 K [3] showed that the tensile yield strength decreased from 610 to 380 MPa with increasing the test temperature from room temperature to 473 K. It is significant that the alloy remained at high strength, exceeding 500 MPa even at a high temperature of 423 K while the elevated temperature strength of the commercial elevated temperature ingot WE54-T6 (Mg-Y-Nd) alloys was 200 MPa at temperatures below 473 K. The RS PM alloy exhibited a large elongation of 300% or more at a wide strain rate range from 10^{-2} to 10^0 s^{-1} . The elongation reached a maximum of 780% at a strain rate of $6 \times 10^{-2} \text{ s}^{-1}$. At the same time, superplastic materials exhibited large in a uniaxial tension [17].

The AMX602 (Mg-6.0Al-0.5Mn-2.0Ca/mass%) and ZAXE1711 (Mg-1.0Zn-7.0Al-1.0Ca-1.0La/mass%) magnesium alloys indented for armor applications were developed collaboratively by the US Army Research Laboratory (ARL) and the Joining and Welding Research Institute (JWRI) of Osaka University [2]. These powders produced by the spinning water atomization process (SWAP) were used as raw initial materials [18,19]. The coarse Mg alloy powders were 1–5 mm size. It was previously verified that the coarse Mg powders of these sizes were noncombustible. The α -Mg grain size of the raw powders was $<0.5 \mu\text{m}$. These raw powders were subjected by compaction and hot extrusion to fabricate the extruded bars. The bar had a cross-section of 24.5 mm $\times$ 40 mm $\times$ 1000 mm.

Jones and Kondoh [2] reported that by SWAP powder preparation, schematically illustrated in Fig. 17.10, noncombustive AMX602 magnesium alloy ingots were melted at 1053 K in the ceramic crucible in the inert gas environment. The effluent molten strickle through the nozzle in the crucible bottom is dispersed by the gas atomizer. Molten metal particles entered in the rotatable water stream on the inner surface of the chamber walls are rapidly crystallized. The chemical compositions of AMX602 alloy granules prepared by SWAP were the following: 6.01Al-0.26Mn-2.09Ca-0.038Si-0.004Cu-0.002Fe-0.007Zn-Mg Bal/mass%. The calcium is necessary because it promotes the noncombustive properties of the magnesium alloys. The impurity content of corrosive elements in magnesium alloys Fe and Cu is controlled to $<0.005\%$. The coarse AMX602 granules prepared by SWAP were 1–5 mm in size with an irregular shape.

Fig. 17.11 shows the optical microstructures of the (A) AMX602 cast ingot, (C) granules, and (B) as-received powders prepared by SWAP. The cast ingot material consists of coarse α -Mg grains of 60–150 μm diameter. As shown in Fig. 17.11C, the powders revealed small dendrite structures that were formed during the rapid solidification of molten Mg alloy droplets. A mean dendrite arm spacing was 0.97 μm . The estimated solidification rate of SWAP Mg alloy powders is

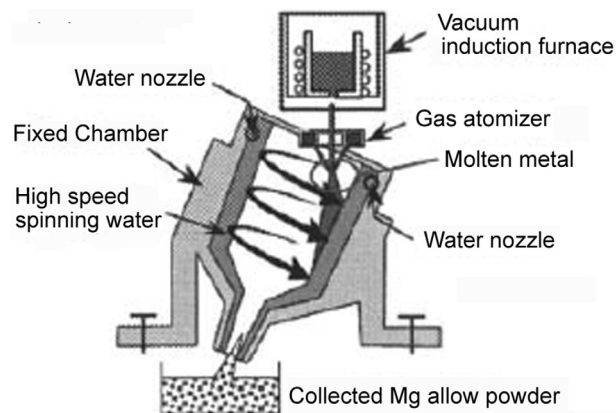


FIG. 17.10 Schematic illustration of spinning water atomization process (SWAP) installation to produce rapidly solidified Mg alloy granules. (Modified from Sakamoto M, Akiyama S, Hagio T, Ogi K. Control of oxidation surface film and suppression of ignition of molten Mg-Ca alloy by Ca addition. *J Jpn Foundry Eng Soc* 1997; 69: 227–233.)

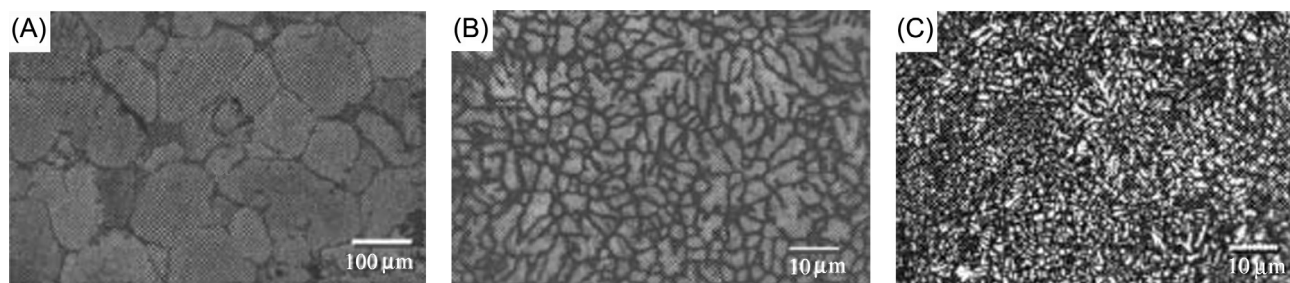


FIG. 17.11 Optical microstructures of AMX602 cast ingot (A), powders (B), and granules (C) prepared by SWAP. (From Jones T, Kondoh K. *Ballistic analysis of new military grade magnesium alloys for armor applications*. In: *Magnesium technology 2011*. TMS (The Minerals, Metals and Materials Society); 2011. p. 425–30.)

$1.1 \times 10^5 \text{ K} \cdot \text{s}^{-1}$. The estimated solidification rate of SWAP Mg alloy granules is $1.1 \times 10^3 \text{ K} \cdot \text{s}^{-1}$. The mean particle size of fine particles is 0.36 μm . The mean particle size of coarse powders is 1.87 μm .

According to data [2], the AMX602 magnesium alloy fabricated by extruding the green compacts of rapidly solidified granules showed α -Mg grains of 0.3–1.1 μm in diameter. Compared to the AMX602 extruded alloys using the cast ingot billets, the UTS and YS of the granule-extruded materials showed an increment of 30%–45%. The 422 MPa UTS and 14.2% elongation of samples are achieved by the preheating temperature at 623 K. Because of advanced powder metallurgy processing and chemical alloying, superior mechanical properties were achieved. The initial ballistic performance results of the Mg alloy AMX602 showed up to a 33% higher ballistic limit compared to the baseline Mg armor alloy AZ31B.

The additive friction stir (AFS) manufacturing process for metallic materials in which the added material can be in the form of powder or solid feedstock was applied by Kandasamy et al. [20] in the study of the mechanical properties of WE43 magnesium deposited by AFS using both gas-atomized powder and solid feed for their comparisons. The process is schematically explained in Fig. 17.12. The process principle is similar to friction stir welding processing, with the exception of the use of a filler feed. The filler metal is forced to flow between the rotating AFS tool and the substrate, whereby the filler material undergoes shear-induced severe plastic deformation, dynamic recrystallization, and deposition while the firmly clamped substrate is moving. The hydrostatic pressure and the heat generated from the adiabatic and frictional processes are crucial in obtaining sound bonding between the substrate and filler metal. The severe plastic deformation imparted into the filler and substrate interface yields a dense, homogenous, fine-grained wrought microstructure. The nominal chemical composition of the magnesium alloy WE43 substrate and the filler metal was Mg-4Y-3Nd-0.5Zr/mass%. The solid filler material was extracted from a rolled plate and used in the as-received and softened (798 K for 8 h) condition. The powder filler specimen sizes were 127 mm $\times$ 25.4 mm $\times$ 6.35 mm.

Kandasamy et al. [20] reported that the tensile strength, yield strength, and ductility of the base metal were $323 \pm 3 \text{ MPa}$, $285 \pm 1 \text{ MPa}$, and $12 \pm 1\%$, respectively. The tensile strength of the deposit made using as-received filler in as-deposited condition was close to 300 MPa, which is 20 MPa lower than the base metal. A maximum tensile strength of 360 MPa and yield strength of 330 MPa with 11% ductility of the powder-deposited specimens were obtained upon aging at 273 K for 24 h. The authors are of the opinion that further optimization studies are needed to realize the full potential of powder-deposited samples.

Magnesium and its alloys are promising materials for biodegradable vascular stents, owing to their relatively low corrosion resistance to human body fluids and their good biocompatibility [21–23]. However, studies have also revealed that the rapid corrosion rate of conventional magnesium alloys causes premature loss of stent mechanical properties. The most

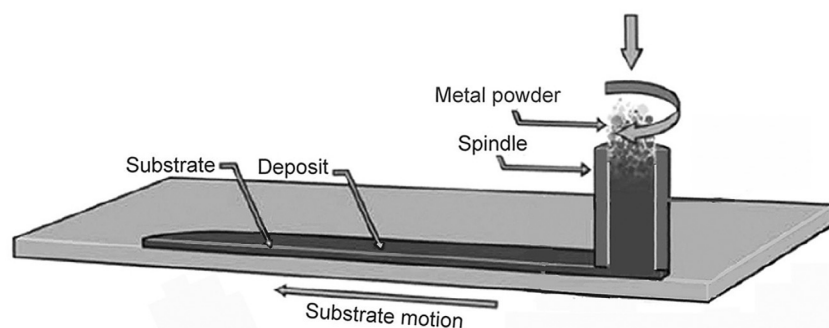


FIG. 17.12 Schematic representation of additive friction stir (AFS) process.

effective way to enhance both the mechanical properties and the corrosion resistance of engineering magnesium alloys is to add specific alloying elements such as rare earth elements [24–27]. However, the toxicity of these alloying elements in a biomaterial is still a controversial issue among biomedical scientists [22,26,28]. The studies by Alvarez-Lopez et al. [29] and Argade et al. [30] reported that in a AZ31 magnesium alloy, the best corrosion behavior in the phosphate-buffer solution could be achieved after extensive grain refinement by equal-channel angular pressing.

Ge et al. [31] concentrated their experimental investigation on a ZM21 magnesium alloy selected with the aim of exploring an alloy system preferentially formed by nontoxic elements, thus preserving the highest levels of biosafety and biocompatibility. It is reported that Ca, Zn, and Mn in fairly low concentrations do not produce harmful effects (these elements are actually essential for the human metabolism) [26,28,32], whereas elements such as Al, Zr, Y, and other RE elements that are used in other commercial Mg alloys to improve strength and corrosion resistance may give unwanted effects when released into the human body at high rates [22,23,26,33].

Wolff M, et al. reported [34] that significant enhancement of mechanical properties of Mg-based alloy biomedical implant parts was realized using 35 wt% of backbone polymer in the binder system of the feedstock.

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